Sec. 14.1

1. Find an equation for and sketch the graph of the level curve of the function

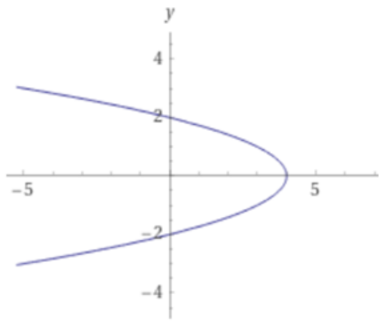
$f(x,y)=\sqrt{x+y^2-3}$ that passes through the point $(3,-1)$.

求函數 $f(x,y)=\sqrt{x+y^2-3}$ 通過點 $(3,-1)$ 的等高線方程式並畫出此等高線。

$z=f(3,-1)=\sqrt{3+(-1)^2-3}=1$

$f(x,y)=1$ is the level at height 1

$\sqrt{x+y^2-3}=1 \Rightarrow x+y^2-3=1 \Rightarrow x+y^2=4$



Sec. 14.3

Calculate first-order partial derivatives $\frac{\partial f}{\partial x}$, $\frac{\partial f}{\partial y}$.

計算一階偏導數

2. $f(x,y)=x\sin(x^2y)$

$$f_x=\frac{\partial}{\partial x}[x\sin(x^2y)]$$

$$=(\frac{\partial}{\partial x}x)\sin(x^2y)+x\frac{\partial}{\partial x}\sin(x^2y)=\sin(x^2y)+x[\cos(x^2y)\frac{\partial}{\partial x}(x^2y)]$$

$$=\sin(x^2y)+x\cos(x^2y)(2xy)=\sin(x^2y)+2x^2y\cos(x^2y)$$

$$f_y=x\frac{\partial}{\partial y}\sin(x^2y)=x\cos(x^2y)\frac{\partial}{\partial y}(x^2y)=x\cos(x^2y)(x^2)=x^3\cos(x^2y)$$

3. $f(x,y)=\ln\sqrt{x^2+y^2}$

$$f(x,y)=\frac{1}{2}\ln(x^2+y^2)$$

$$f_x=\frac{1}{2}\frac{\partial}{\partial x}\ln(x^2+y^2)=\frac{1}{2}\frac{1}{x^2+y^2}\frac{\partial}{\partial x}(x^2+y^2)=\frac{1}{2}\frac{2x}{x^2+y^2}=\frac{x}{x^2+y^2}$$

$$f_y=\frac{1}{2}\frac{\partial}{\partial y}\ln(x^2+y^2)=\frac{1}{2}\frac{1}{x^2+y^2}\frac{\partial}{\partial y}(x^2+y^2)=\frac{1}{2}\frac{2y}{x^2+y^2}=\frac{y}{x^2+y^2}$$

....

Calculate second-order partial derivatives $f_{xx}, f_{xy}, f_{yx}, f_{yy}$.

計算二階偏導數

4. $f(x,y)=e^{xy}$

$$f_x=\frac{\partial}{\partial x}e^{xy}=e^{xy}\frac{\partial}{\partial x}(xy)=ye^{xy}\text{，}\ f_y=\frac{\partial}{\partial y}e^{xy}=e^{xy}\frac{\partial}{\partial y}(xy)=xe^{xy}$$

$$f_{xx}=\frac{\partial}{\partial x}f_x=y\frac{\partial}{\partial x}e^{xy}=y[e^{xy}\frac{\partial}{\partial x}(xy)]=y(ye^{xy})=y^2e^{xy}$$

$$f_{xy}=\frac{\partial}{\partial y}f_x=\frac{\partial}{\partial y}[ye^{xy}]=(\frac{\partial}{\partial y}y)e^{xy}+y(\frac{\partial}{\partial y}e^{xy})=e^{xy}+y(xe^{xy})=e^{xy}(1+xy)$$

$$f_{yx}=\frac{\partial}{\partial x}f_y=\frac{\partial}{\partial x}[xe^{xy}]=(\frac{\partial}{\partial x}x)e^{xy}+x(\frac{\partial}{\partial x}e^{xy})=e^{xy}+x(ye^{xy})=e^{xy}(1+xy)$$

$$f_{yy}=\frac{\partial}{\partial y}f_y=x\frac{\partial}{\partial x}e^{xy}=x[e^{xy}\frac{\partial}{\partial y}(xy)]=x(xe^{xy})=x^2e^{xy}$$

....

5. $f(x,y)=\tan^{-1}\dfrac{y}{x}$

$$f_x=\frac{\partial}{\partial x}\tan^{-1}\frac{y}{x}=\frac{1}{1+(\frac{y}{x})^2}\frac{\partial}{\partial x}\frac{y}{x}=\frac{1}{1+(\frac{y}{x})^2}\frac{-y}{x^2}=\frac{-y}{x^2+y^2}$$

$$f_y=\frac{\partial}{\partial y}\tan^{-1}\frac{y}{x}=\frac{1}{1+(\frac{y}{x})^2}\frac{\partial}{\partial y}\frac{y}{x}=\frac{1}{1+(\frac{y}{x})^2}\frac{1}{x}=\frac{x}{x^2+y^2}$$

$$f_{xx}=\frac{\partial}{\partial x}f_x=-y\frac{\partial}{\partial x}\frac{1}{x^2+y^2}=-y\frac{-2x}{(x^2+y^2)^2}=\frac{2xy}{(x^2+y^2)^2}$$

$$f_{xy}=\frac{\partial}{\partial y}f_x=-\frac{\partial}{\partial y}\frac{y}{x^2+y^2}=-\frac{1(x^2+y^2)-y(2y)}{(x^2+y^2)^2}=-\frac{x^2-y^2}{(x^2+y^2)^2}=\frac{y^2-x^2}{(x^2+y^2)^2}$$

$$f_{yx}=\frac{\partial}{\partial x}f_y=\frac{\partial}{\partial x}\frac{x}{x^2+y^2}=\frac{1(x^2+y^2)-x(2x)}{(x^2+y^2)^2}=\frac{y^2-x^2}{(x^2+y^2)^2}$$

$$f_{yy}=\frac{\partial}{\partial y}f_y=x\frac{\partial}{\partial y}\frac{1}{x^2+y^2}=x\frac{-2y}{(x^2+y^2)^2}=\frac{-2xy}{(x^2+y^2)^2}$$

Differentiating Implldtly 偏隱微分

#6. Find the value of $\dfrac{\partial z}{\partial x}$ at the point $(1,1,1)$ if the equation $xy+z^3x-2yz=0$ defmes z as a function of the two independent variables x and y and the partial derivative exists.

若 z 是 x 及 y 的函數且滿足方程式 $xy+z^3x-2yz=0$，求 $\dfrac{\partial z}{\partial x}$ 在點 $(1,1,1)$ 的值。

解：

$\dfrac{\partial}{\partial x}(xy+z^3x-2yz)=\dfrac{\partial}{\partial x}0$ ，兩邊微分

$y\dfrac{\partial}{\partial x}x+\dfrac{\partial}{\partial x}(z^3x)-2y\dfrac{\partial}{\partial x}z=0$ ，視 x 為變數而 y 為常數，常數倍先提出來

$y+(3z^2\dfrac{\partial z}{\partial x}\cdot x+z^3)-2y\dfrac{\partial z}{\partial x}=0$ ，積法則

$(3xz^2-2y)\dfrac{\partial z}{\partial x}=-(y+z^3)\Rightarrow\dfrac{\partial z}{\partial x}=\dfrac{y+z^3}{2y-3xz^2}$ ，含 $\dfrac{\partial z}{\partial x}$ 的項在左邊

Sec. 14.4

1. Use the Chain Rule to find the values of $\frac{dw}{dt}$

if $w=\ln(x^2+y^2+z^2)$, $x=\cos t$, $y=\sin t$, $z=4\sqrt{t}$.

解：

$$\frac{dw}{dt}=\frac{\partial w}{\partial x}\frac{dx}{dt}+\frac{\partial w}{\partial y}\frac{dy}{dt}+\frac{\partial w}{\partial z}\frac{dz}{dt}$$

$$=\frac{2x}{x^2+y^2+z^2}(-\sin t)+\frac{2y}{x^2+y^2+z^2}(\cos t)+\frac{2z}{x^2+y^2+z^2}(\frac{2}{\sqrt{t}})$$

#2. Use the Chain Rule to find the values of $\frac{\partial z}{\partial u}$ and $\frac{\partial z}{\partial v}$

if $z=4e^x\ln y$, $x=\ln(u\cos v)$, $y=u\sin v$.

解：

$$x=\ln(u\cos v)=\ln u+\ln\cos v$$

$$\frac{\partial z}{\partial u}=\frac{\partial z}{\partial x}\frac{\partial x}{\partial u}+\frac{\partial z}{\partial y}\frac{\partial y}{\partial u}=(4e^x\ln y)(\frac{1}{u})+(\frac{4e^x}{y})(\sin v)$$

$$\frac{\partial z}{\partial v}=\frac{\partial z}{\partial x}\frac{\partial x}{\partial v}+\frac{\partial z}{\partial y}\frac{\partial y}{\partial v}=(4e^x\ln y)(\frac{-\sin v}{\cos v})+(\frac{4e^x}{y})(u\cos v)$$

---

3. Assuming that the equation $xe^y+\sin xy+y-\ln 2=0$ define y as a differentiable function of x, use Theorem 8 to find the value of $\frac{dy}{dx}$.

解：

$$F(x,y)=xe^y+\sin xy+y-\ln 2$$

$F_x=e^y+y\cos xy$ ， $F_y=xe^y+x\cos xy+1$

$$\frac{dy}{dx}=-\frac{F_x}{F_y}=-\frac{e^y+y\cos xy}{xe^y+x\cos xy+1}$$

#4. Assuming that the equation $\sin(x+y)+\sin(y+z)+\sin(z+x)=0$ define z as a differentiable function of x and y,

use Theorem 8 to find the values of $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.

解：

$$F(x,y,z)=\sin(x+y)+\sin(y+z)+\sin(z+x)$$

$$F_x=\cos(x+y)+\cos(z+x) \text{，} F_y=\cos(x+y)+\cos(y+z) \text{，} F_z=\cos(y+z)+\cos(z+x)$$

$$\frac{\partial z}{\partial x}=-\frac{F_x}{F_z}=-\frac{\cos(x+y)+\cos(z+x)}{\cos(y+z)+\cos(z+x)} \quad , \quad \frac{\partial z}{\partial y}=-\frac{F_y}{F_z}=-\frac{\cos(x+y)+\cos(y+z)}{\cos(y+z)+\cos(z+x)}$$

---

5. Assume that $w=f(s^3+t^2)$ and $f'(x)=e^x$. Find $\frac{\partial w}{\partial t}$ and $\frac{\partial w}{\partial s}$.

解：

$$x=s^3+t^2$$

$$\frac{\partial w}{\partial t}=\frac{dw}{dx}\frac{\partial x}{\partial t}=f'(x)\frac{\partial x}{\partial t}=e^x(2t) \text{，} \frac{\partial w}{\partial s}=\frac{dw}{dx}\frac{\partial x}{\partial s}=f'(x)\frac{\partial x}{\partial s}=e^x(3s^2)$$

#6. Assume that $w=f(ts^2,\frac{s}{t})$ and $\frac{\partial f}{\partial x}(x,y)=xy,\ \frac{\partial f}{\partial y}(x,y)=\frac{x^2}{2}$. Find $\frac{\partial w}{\partial t}$ and $\frac{\partial w}{\partial s}$.

解：

$$x=ts^2,\ y=\frac{s}{t}$$

$$\frac{\partial w}{\partial t}=\frac{\partial w}{\partial x}\frac{\partial x}{\partial t}+\frac{\partial w}{\partial y}\frac{\partial y}{\partial t}=(xy)(s^2)+(\frac{x^2}{2})(-\frac{s}{t^2})$$

$$\frac{\partial w}{\partial s}=\frac{\partial w}{\partial x}\frac{\partial x}{\partial s}+\frac{\partial w}{\partial y}\frac{\partial y}{\partial s}=(xy)(2st)+(\frac{x^2}{2})(\frac{1}{t})$$

#43. If $f(u,v,w)$ is differentiable and $u=x-y$, $v=y-z$, and $w=z-x$,

show that $\frac{\partial f}{\partial x}+\frac{\partial f}{\partial y}+\frac{\partial f}{\partial z}=0$.

解：

$$\frac{\partial f}{\partial x}=\frac{\partial f}{\partial u}\frac{\partial u}{\partial x}+\frac{\partial f}{\partial v}\frac{\partial v}{\partial x}+\frac{\partial f}{\partial w}\frac{\partial w}{\partial x}=\frac{\partial f}{\partial u}(1)+\frac{\partial f}{\partial v}(0)+\frac{\partial f}{\partial w}(-1)=\frac{\partial f}{\partial u}-\frac{\partial f}{\partial w}\cdots(1)$$

$$\frac{\partial f}{\partial y}=\frac{\partial f}{\partial u}\frac{\partial u}{\partial y}+\frac{\partial f}{\partial v}\frac{\partial v}{\partial y}+\frac{\partial f}{\partial w}\frac{\partial w}{\partial y}=\frac{\partial f}{\partial u}(-1)+\frac{\partial f}{\partial v}(1)+\frac{\partial f}{\partial w}(0)=-\frac{\partial f}{\partial u}+\frac{\partial f}{\partial v}\cdots(2)$$

$$\frac{\partial f}{\partial z}=\frac{\partial f}{\partial u}\frac{\partial u}{\partial z}+\frac{\partial f}{\partial v}\frac{\partial v}{\partial z}+\frac{\partial f}{\partial w}\frac{\partial w}{\partial z}=\frac{\partial f}{\partial u}(0)+\frac{\partial f}{\partial v}(-1)+\frac{\partial f}{\partial w}(1)=-\frac{\partial f}{\partial v}+\frac{\partial f}{\partial w}\cdots(3)$$

$$\frac{\partial f}{\partial x}+\frac{\partial f}{\partial y}+\frac{\partial f}{\partial z}=(1)+(2)+(3)=0$$

Sec. 14.5

Calculating Gradients

In Exercises 1–6, find the gradient of the function at the given point. Then sketch the gradient together with the level curve that passes through the point.

5. $f(x,y)=\sqrt{2x+3y}$, $(-1,2)$

$$f_x=\frac{2}{2\sqrt{2x+3y}}=\frac{1}{\sqrt{2x+3y}},\ f_y=\frac{3}{2\sqrt{2x+3y}}$$

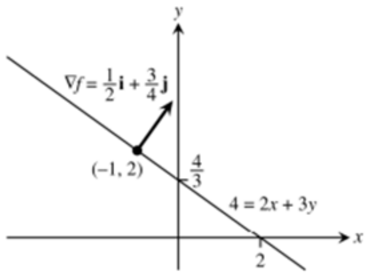
$$f_x(-1,2)=\frac{1}{2},\ f_y(-1,2)=\frac{3}{4}$$

在點$(-1,2)$的梯度$\nabla f(-1,2)=<f_x(-1,2),f_y(-1,2)>=<\frac{1}{2},\frac{3}{4}>$

$$f(-1,2)=\sqrt{-2+6}=2$$

在點$(-1,2)$的等高線$f(x,y)=2\Rightarrow$等高線方程式$\sqrt{2x+3y}=2$

兩邊平方，等高線方程式$2x+3y=4$是一直線



Finding Directional Derivatives

In Exercises 11–18, find the derivative of the function at P_0 in the direction of **u**.

13. $g(x, y) = \dfrac{x - y}{xy + 2}$, $P_0(1, -1)$, $\mathbf{u} = 12\mathbf{i} + 5\mathbf{j}$

不是單位向量，要化為單位向量$\mathbf{u} = \dfrac{\mathbf{v}}{|\mathbf{v}|} = \dfrac{<12,5>}{|<12,5>|} = \dfrac{<12,5>}{\sqrt{12^2+5^2}} = <\dfrac{12}{13}, \dfrac{5}{13}>$

$$g_x = \frac{1(xy+2)-(x-y)y}{(xy+2)^2} = \frac{2+y^2}{(xy+2)^2},\quad g_y = \frac{-1(xy+2)-(x-y)x}{(xy+2)^2} = -\frac{2+x^2}{(xy+2)^2}$$

$g_x(1,-1) = 3,\ g_y(1,-1) = -3$

在點$(1,-1)$的梯度$\nabla g(1,-1) = <g_x(1,-1), g_y(1,-1)> = <3,-3>$

g在點$(1,-1)$於方向$\mathbf{v} = 12\mathbf{i}+5\mathbf{j}$的方向導數

$$(D_{\mathbf{u}}g)_{(1,-1)} = (\nabla g)_{(1,-1)} \cdot \mathbf{u} = <3,-3> \cdot <\frac{12}{13}, \frac{5}{13}> = \frac{36}{13} - \frac{15}{13} = \frac{21}{13}$$

In Exercises 19–24, find the directions in which the functions increase and decrease most rapidly at P_0. Then find the derivatives of the functions in these directions.

20. $f(x, y) = x^2y + e^{xy}\sin y$, $P_0(1, 0)$

梯度$\nabla f = <2xy + ye^{xy}\sin y, x^2 + xe^{xy}\sin y + e^{xy}\cos y>$

在點$P_0(1,0)$的梯度$\nabla f(1,0) = <0,2>$

$|\nabla f(1,0)| = \sqrt{0^2+2^2} = 2$

(a) 與梯度$\nabla f(1,0) = <0,2>$同向時，導數$(D_{\mathbf{u}}f)_{P_0}$最大

即單位方向向量$\mathbf{u} = <0,1>$，$(D_{\mathbf{u}}f)_{P_0} = |\nabla f(1,0)| = 2$最大

(b) 與梯度$\nabla f(1,0) = <0,2>$反向時，$(D_{\mathbf{u}}f)_{P_0}$最小

即單位方向向量$\mathbf{u} = <0,-1>$，$(D_{\mathbf{u}}f)_{P_0} = -|\nabla f(1,0)| = -2$最小

Theory and Examples

29. Let $f(x, y) = x^2 - xy + y^2 - y$. Find the directions $\mathbf{u}$ and the values of $D_{\mathbf{u}} f(1, -1)$ for which

a. $D_{\mathbf{u}} f(1, -1)$ is largest **b.** $D_{\mathbf{u}} f(1, -1)$ is smallest

c. $D_{\mathbf{u}} f(1, -1) = 0$

梯度 $\nabla f =< 2x - y, -x + 2y - 1 >$，在點 $P_0(1,-1)$ 的梯度 $\nabla f(1,-1) =< 3,-4 >$

$|\nabla f(1,-1)| = \sqrt{3^2 + (-4)^2} = \sqrt{25} = 5$

(b) 與梯度 $\nabla f(1,-1) =< 3,-4 >$ 同向時，導數 $(D_{\mathbf{u}} f)_{P_0}$ 最大

即單位方向向量 $\mathbf{u} =< \frac{3}{5}, \frac{-4}{5} >$，$(D_{\mathbf{u}} f)_{P_0} = |\nabla f(1,-1)| = 5$ 最大

(b) 與梯度 $\nabla f(1,-1) =< 3,-4 >$ 反向時，$(D_{\mathbf{u}} f)_{P_0}$ 最小

即單位方向向量 $\mathbf{u} =< \frac{-3}{5}, \frac{4}{5} >$，$(D_{\mathbf{u}} f)_{P_0} = -|\nabla f(1,-1)| = -5$ 最小

(c) 與梯度 $\nabla f(1,-1) =< 3,-4 >$ 垂直時，導數 $(D_{\mathbf{u}} f)_{P_0} = 0$

即單位方向向量 $\mathbf{u} =< \frac{4}{5}, \frac{3}{5} >$ 或 $\mathbf{u} =< \frac{-4}{5}, \frac{-3}{5} >$，

$$(D_{\mathbf{u}} f)_{P_0} =< \frac{3}{5}, \frac{-4}{5} > \bullet < \frac{4}{5}, \frac{3}{5} > = \frac{12}{25} + \frac{-12}{25} = 0$$

$$(D_{\mathbf{u}} f)_{P_0} =< \frac{3}{5}, \frac{-4}{5} > \bullet < \frac{-4}{5}, \frac{-3}{5} > = \frac{-12}{25} + \frac{12}{25} = 0$$

....

Sec. 14.6

Tangent Planes and Normal Lines to Surfaces

In Exercises 1–8, find equations for the

(a) tangent plane and

(b) normal line at the point P_0 on the given surface.

5. $\cos \pi x - x^2 y + e^{xz} + yz = 4, \quad P_0(0, 1, 2)$

$$f(x, y, z) = \cos \pi x - x^2 y + e^{xz} + yz$$

$$f_x = -\pi \sin \pi x - 2xy + ze^{xz},\ f_y = -x^2 + z,\ f_z = xe^{xz} + y$$

$$f_x(0,1,2) = 2,\ f_y(0,1,2) = 2,\ f_z(0,1,2) = 1$$

....

求在點$(0,1,2)$切平面方程式，方法一

直接帶入切平面方程式公式$f_x(P_0)\cdot(x-x_0)+f_y(P_0)\cdot(y-y_0)+f_z(P_0)\cdot(z-z_0)=0$

得到$2(x-0)+2(y-1)+1(z-2)=0$ 或 $2x+2y+z-4=0$

………………………………

求在點$(0,1,2)$切平面方程式，方法二

設(x,y,z)是切平面上任一點，則$<x-0,y-1,z-2>$是切平面上的向量，

法向量＝梯度$\nabla f(0,1,2)=<f_x,f_y,f_z>=<2,2,1>$

切平面與梯度向量垂直，

所以$<x-0,y-1,z-2>\bullet<2,2,1>=0$

得到切平面方程式$2(x-0)+2(y-1)+1(z-2)=0$ 或 $2x+2y+z-4=0$

………………………………

設(x,y,z)是法線上任一點，

在點$(0,1,2)$的法線方程式$\dfrac{x-0}{2}=\dfrac{y-1}{2}=\dfrac{z-2}{1}$

或 $\dfrac{x-0}{2}=\dfrac{y-1}{2}=\dfrac{z-2}{1}=t$，$x=2t,\ y=1+2t,\ z=2+t$

……………………………………………………………………………………………………

In Exercises 9–12, find an equation for the plane that is tangent to the given surface at the given point.

11. $z=\sqrt{y-x}$, $(1,2,1)$

$f(x,y)=\sqrt{y-x}$，$z_0=f(1,2)=\sqrt{2-1}=1$

$f_x=-\dfrac{1}{2\sqrt{y-x}}, f_y=\dfrac{1}{2\sqrt{y-x}}$，$f_x(1,2)=-\dfrac{1}{2}, f_y(1,2)=\dfrac{1}{2}$

………………………………

求在點$(1,2,1)$切平面方程式，方法一

直接帶入切平面方程式公式 $f_x(P_0)\cdot(x-x_0)+f_y(P_0)\cdot(y-y_0)-(z-z_0)=0$

得到$-\dfrac{1}{2}(x-1)+\dfrac{1}{2}(y-2)-(z-1)=0$ 或 $x-y+2z-1=0$

………………………………

求切平面方程式，方法二

設(x,y,z)是切平面上任一點，則$<x-1,y-2,z-1>$是切平面上的向量，

法向量＝$<f_x(1,2),f_y(1,2),-1>=<-\dfrac{1}{2},\dfrac{1}{2},-1>$

切平面與梯度向量垂直，

所以$<x-1, y-2, z-1> \bullet <-\frac{1}{2}, \frac{1}{2}, -1> = 0$

得到切平面方程式$-\frac{1}{2}(x-1)+\frac{1}{2}(y-2)-(z-1)=0$ 或 $x-y+2z-1=0$

Finding Linearizations

In Exercises 25–30, find the linearization $L(x, y)$ of the function at each point.

30. $f(x, y) = e^{2y-x}$ at **b.** $(1, 2)$

解：已知$f(1,2)=e^{4-1}=e^3$

$f_x=-e^{2y-x}$，$f_y=2e^{2y-x}$；$f_x(1,2)=-e^3$，$f_y(1,2)=2e^3$

f在(x_0, y_0)的線性化$L(x,y)=f(x_0,y_0)+f_x(x_0,y_0)\cdot(x-x_0)+f_y(x_0,y_0)\cdot(y-y_0)$

f在點$(1,2)$的線性化$L(x,y)=f(1,2)+f_x(1,2)\cdot(x-1)+f_y(1,2)\cdot(y-2)$

$L(x,y)=e^3-e^3\cdot(x-1)+2e^3\cdot(y-2)=-e^3x+2e^3y-2e^3$

Estimating Error; Sensitivity to Change

50. Estimating volume of a cylinder About how accurately may $V = \pi r^2 h$ be calculated from measurements of r and h that are in error by 1%?

體積$V=\pi r^2 h$，$V_r=2\pi rh$，$V_h=\pi r^2$，題意知 $\left|\frac{dr}{r}\right|\le 0.01$，$\left|\frac{dh}{h}\right|\le 0.01$

體積的變化量$\Delta V \approx dV = V_r\cdot dr+V_h\cdot dh = 2\pi rh\cdot dr+\pi r^2\cdot dh$

體積的可能相對誤差

$$\left|\frac{dV}{V}\right|=\left|\frac{2\pi rh\cdot dr+\pi r^2\cdot dh}{\pi r^2 h}\right|\le\left|\frac{2dr}{r}\right|+\left|\frac{dh}{h}\right|=2\left|\frac{dr}{r}\right|+\left|\frac{dh}{h}\right|\le 2(0.01)+0.01=0.03$$

體積的最大可能百分比誤差 $\left|\frac{dV}{V}\right|100\% \le (0.03)\cdot 100\% = 3\%$

Sec. 14.7

練習作業 Homework：13,17,20

習題 13：求 $f(x,y)=x^3-y^3-2xy+6$ 的局部極值。　　沒有判定就沒有分數

解：

$f_x(x,y)=3x^2-2y$，$f_y(x,y)=-3y^2-2x$

求臨界點 $\begin{cases} f_x(x,y)=0 \\ f_y(x,y)=0 \end{cases} \Rightarrow \begin{cases} 3x^2-2y=0 \\ -3y^2-2x=0 \end{cases} \Rightarrow \begin{cases} y=\frac{3}{2}x^2 \\ x=-\frac{3}{2}y^2 \end{cases}$

將第二式代入第一式得到

$y=\frac{3}{2}(-\frac{3}{2}y^2)^2 \Rightarrow y=\frac{3^3}{2^3}y^4 \Rightarrow 8y=27y^4 \Rightarrow y(27y^3-8)=0 \Rightarrow y(3y-2)(9y^2+6y+4)=0$

得到 $y=0$，$y=\frac{2}{3}$；$y=0$ 時 $x=0$；$y=\frac{2}{3}$ 時 $x=-\frac{2}{3}$；$(0,0), (-\frac{2}{3},\frac{2}{3})$ 為臨界點。

$f_{xx}(x,y)=6x$，$f_{xy}(x,y)=-2$，$f_{yy}(x,y)=-6y$

判別式 $D(x,y)=(6x)(-6y)-(-2)^2=-36xy-4$

臨界點	$D(x,y)$	$f_{xx}(x,y)$	$f(x,y)$
$(0,0)$	$-4<0$	×	$(0,0,6)$ 鞍點
$(-\frac{2}{3},\frac{2}{3})$	$12>0$	$-4<0$	$\frac{120}{27}$ 相對極大值

習題 17：求 $f(x,y)=x^3+3xy^2-15x+y^3-15y$ 的局部極值。　　沒有判定就沒有分數

解：

$f_x(x,y)=3x^2+3y^2-15$，$f_y(x,y)=6xy+3y^2-15$

求臨界點 $\begin{cases} f_x(x,y)=0 \\ f_y(x,y)=0 \end{cases} \Rightarrow \begin{cases} 3x^2+3y^2-15=0 \\ 6xy+3y^2-15=0 \end{cases} \Rightarrow \begin{cases} x^2+y^2-5=0 \\ 2xy+y^2-5=0 \end{cases}$

第一式減第二式得到

$x^2-2xy=0 \Rightarrow x(x-2y)=0 \Rightarrow x=0,\ x=2y$

$x=0$ 時 $y=\pm\sqrt{5}$；$x=2y$ 時代入第一式 $5y^2=5$；$y=\pm1$，$(0,\pm\sqrt{5}), (2,1), (-2,-1)$ 為臨界點。

二階偏導數 $f_{xx}(x,y)=6x$，$f_{xy}(x,y)=6y$，$f_{yy}(x,y)=6x+6y$

判別式 $D(x,y)=(6x)(6x+6y)-(6y)^2$

臨界點	$D(x,y)$	$f_{xx}(x,y)$	$f(x,y)$
$(0,\sqrt{5})$	$-180<0$	×	鞍點
$(0,-\sqrt{5})$	$-180<0$	×	鞍點
$(2,1)$	$180>0$	$12>0$	-30 相對極小值
$(-2,-1)$	$180>0$	$-12<0$	30 相對極大值

習題 19：求 $f(x,y)=x^4+y^4+4xy$ 的局部極值。　　沒有判定就沒有分數

解：

$f_x(x,y)=4x^3+4y$，$f_y(x,y)=4y^3+4x$

求臨界點 $\begin{cases} f_x(x,y)=0 \\ f_y(x,y)=0 \end{cases} \Rightarrow \begin{cases} 4x^3+4y=0 \\ 4y^3+4x=0 \end{cases} \Rightarrow \begin{cases} y=-x^3 \\ x=-y^3 \end{cases}$

將第一式代入第二式得到

$x=x^9 \Rightarrow x(x^8-1)=x(x-1)(x+1)(x^2+1)(x^4+1)=0 \Rightarrow x=-1,0,1$

$x=-1$ 時 $y=1$；$x=0$ 時 $y=0$；$x=1$ 時 $y=-1$，$(1,-1),(0,0),(-1,1)$ 為臨界點。

二階偏導數 $f_{xx}(x,y)=12x^2$，$f_{xy}(x,y)=4$，$f_{yy}(x,y)=12y^2$

判別式 $D(x,y)=(12x^2)(12y^2)-(4)^2=144x^2y^2-16$

臨界點	$D(x,y)$	$f_{xx}(x,y)$	$f(x,y)$
$(1,-1)$	$128>0$	$12>0$	-2 相對極小值
$(0,0)$	$-16<0$	$\times$	鞍點
$(-1,1)$	$128>0$	$12>0$	-2 相對極小值

Sec. 14.8

練習作業 Homework：3,5,23

3. Maximum on a line Find the maximum value of $f(x,y)=49-x^2-y^2$ on the line $x+3y=10$.

Step 1，$\begin{cases} f_x=\lambda g_x \\ f_y=\lambda g_y \\ g(x,y)=0 \end{cases}$，$\begin{cases} -2x=\lambda\cdot 1\cdots(1) \\ -2y=\lambda\cdot 3\cdots(2) \\ x+3y=10\cdots(3) \end{cases}$，解聯立方程組

$\frac{(1)}{(2)} \Rightarrow \frac{-2x}{-2y}=\frac{\lambda\cdot 1}{\lambda\cdot 3} \Rightarrow \frac{x}{y}=\frac{1}{3} \Rightarrow y=3x$　代入第 3 式得

$x+9x=10 \Rightarrow 10x=10 \Rightarrow x=1,\ y=3$

Step 2　故有極大值 $f(1,3)=49-1^2-3^2=39$，

(可以讓 $x\to\infty, y\to-\infty$ 一正一負，仍滿足 $x+3y=10$，

故 $f(x,y)=49-x^2-y^2\to-\infty$ 不會有極小值)

5. Constrained minimum Find the points on the curve $xy^2 = 54$ nearest the origin.

設(x,y)是曲線xy^2=54上一點，點(x,y)到點$(0,0)$的距離$d=\sqrt{x^2+y^2}$

題意求距離$d=\sqrt{x^2+y^2}$最短的點(x,y)

即求$f(x,y)=x^2+y^2$在限制條件xy^2=54下最小值的點

Step 1，$\begin{cases} f_x=\lambda g_x \\ f_y=\lambda g_y \\ g(x,y)=0 \end{cases}$，$\begin{cases} 2x=\lambda\cdot y^2\cdots(1) \\ 2y=\lambda\cdot 2xy\cdots(2) \\ xy^2=54\cdots(3) \end{cases}$，解聯立方程組

$\frac{(1)}{(2)}\Rightarrow\frac{2x}{2y}=\frac{\lambda\cdot y^2}{\lambda\cdot 2xy}\Rightarrow\frac{x}{y}=\frac{y}{2x}\Rightarrow y^2=2x^2$ 代入第 3 式得

$x(2x^2)=54\Rightarrow x^3=27\Rightarrow x=3,\ y=\pm3\sqrt{2}$

Step 2 故有極小值$f(3,3\sqrt{2})=f(3,-3\sqrt{2})=27$，

(可以讓$x\to0,y\to\infty$仍滿足xy^2=54，$f(x,y)=x^2+y^2$不會有極大值)

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23. Extrema on a sphere Find the maximum and minimum values of

$$f(x,y,z)=x-2y+5z$$

on the sphere $x^2+y^2+z^2=30$.

Step 1，$\begin{cases} f_x=\lambda g_x \\ f_y=\lambda g_y \\ f_z=\lambda g_z \\ g(x,y,z)=0 \end{cases}$，$\begin{cases} 1=\lambda\cdot 2x\cdots(1) \\ -2=\lambda\cdot 2y\cdots(2) \\ 5=\lambda\cdot 2z\cdots(3) \\ x^2+y^2+z^2=30\cdots(4) \end{cases}$，解聯立方程組

$\frac{(1)}{(2)}\Rightarrow\frac{1}{-2}=\frac{\lambda\cdot 2x}{\lambda\cdot 2y}\Rightarrow\frac{1}{-2}=\frac{x}{y}\Rightarrow y=-2x$

$\frac{(1)}{(3)}\Rightarrow\frac{1}{5}=\frac{\lambda\cdot 2x}{\lambda\cdot 2z}\Rightarrow\frac{1}{5}=\frac{x}{z}\Rightarrow z=5x$

代入第 4 式得 $x^2+(-2x)^2+(5x)^2=30\Rightarrow 30x^2=30\Rightarrow x=\pm1$

Step 2，

(x,y,z)	$(1,-2,5)$	$(-1,2,-5)$
$f(x,y)=x-2y+5z$	30	-30

故有極小值-30，極大值30
